SAMPLE - NOT ISSUED. No Engagement Terms are incorporated and no commitment of any kind is made. Issued reports carry the governing terms version here.

Order identity

Field	Value
Client	Acme Labs
Order id	LQI-2026-Troesch-15-2026-08-29-T3
Reference	LQI-2026-Troesch-15
Problem	A plasma-confinement model.
Stated equation & data	y'' = 15*sinh(15*y), y(0) =0, y(1) = 1
Ordered (date in)	2026-08-29
Delivered (date out)	2026-08-29
Service tier	Tier 3: Certified Premium

VERIFICATION REPORT

Class: ODE boundary-value problem | Engine: SIRPY native — piecewise power series (35 pieces) | Generated by SIRPY 0.59.15. Companion / handoff: LQI-2026-Troesch-15. *Luminary Quantum Institute, LLC — computational verification report · format v1.2*

SAMPLE — NOT ISSUED. No Engagement Terms are incorporated and no commitment of any kind is made. Issued reports carry the governing terms version here.

Prepared for	Acme Labs
Reference	LQI-2026-Troesch-15
Date	2026-08-29
Terms	SAMPLE - not issued
Artifact SHA-256	ce472044612b41fc 56e71cb744ee6f1e 92d99cd9303a4b79 c1f3dfbb1ac3853c

1. Problem

A plasma-confinement model. y'' = 15*sinh(15*y), y(0) =0, y(1) = 1 Conditions (derived from the data): y(0) = 0; y'(0) = free (shooting unknown); y(1) = 1. Stated over the interval [0, 1]. Every statement in this report is bound to the solution artifact identified in section 6.

2. Finding

VERIFIED

A solution was obtained and its agreement with the stated equations and data was measured against the acceptance criterion stated in section 3. Internal status code: solved+verified

3. Measurements

Acceptance criterion for VERIFIED: maximum RELATIVE residual below 1e-6 of the equation's own scale (equation scale at the worst point on this run: 2.452e+07), and boundary/initial data within 1e-6 of their scale, sampled at 1387 interior points chosen OFF the interface nodes — off-node sampling is what makes small residuals meaningful for a piecewise method. Sample counts on this report: 1387 verification points (off the seams); the artifact carries its own dense-sample and scan counts -- every set is generated from this same run, so the counts reconcile by construction.

> How to read these measurements. The residual measures how well this answer fits your equation AS STATED: it bounds backward error (this answer solves a nearby problem exactly), not distance from the true answer -- only the independent cross-check, where present, measures that. A clean-through line, where shown, maps where the residual stays under threshold. Characteristic values pin this report to one specific trajectory. The artifact hash ties every claim to one exact file; re- saving the file changes its bytes and fails the check BY DESIGN. Not covered, by design: whether your equation models your system; agreement with other solvers (a second opinion, not a reproduction); solver tolerance as an accuracy claim; differences within the stated tolerance. Nonlinear problems may hold multiple valid solutions; this report covers the one delivered.

The relative residual is pointwise: at each sample the residual is divided by max(1, the local magnitude of the equation's terms) and the worst point is reported. When that worst point sits where the local magnitude is at most 1, the absolute and relative figures coincide.

Equation residual (absolute)	1.588e-03
Equation residual (relative)	2.984e-10
Boundary data residual	5.845e-10
Interface continuity (largest jump)	2.349e-12
Interface nodes measured	34
Interface metric	largest one-sided jump across any piece boundary, over the solution and its derivatives up to order n-1, each side evaluated from its own piece
Solution pieces	35
Independent cross-check (exact energy quadrature)	5.343e-11 relative deviation between the interval length this trajectory implies and the actual interval -- an EXACT relation, so it bounds FORWARD error, not merely backward error

Characteristic values. These bind the report to one specific trajectory — any two functions can share a residual figure, but only one has these values — and each is independently checkable from the artifact. The endpoint value is the state delivered at the end of the solved interval.

y(0) ; y(1)	0 ; 1
max abs(y)	1 at x = 1 [computed from the delivered series representation]
y'(0)	2.44451303e-06

> Residuals are obtained by substituting the computed solution back into the equations as stated. They measure agreement with those equations; they do not assess whether the equations model the intended system. A small residual bounds the BACKWARD error -- this function solves a nearby problem exactly. Near an ill-conditioned point (a fold, a resonance) the FORWARD error can exceed it; the independent cross- check in this section, where present, is the measurement that addresses that gap.

3b. Computed observations

• y is non-decreasing across the 201-point observation grid of the interval

(shape claims use this coarse grid; the residual verification uses its own far denser sample set -- two sets, one run).

> Observations are computed properties of the certified solution, each checkable from the artifact. Their significance for the intended application remains the client's determination.

3d. Solver notices (plain language)

• [NOTE] reconciliation: a raw run notice referenced a stiff/delegated step;

this run is native and no delegation occurred - the raw notice was superseded by this reconciliation check and logged for engineering review; nothing in sections 3-4 depends on it.

• [NOTE] Some solution pieces were stretched past the range where they are

trustworthy; raising partition= or iterations= shrinks the pieces. On a VERIFIED report this is disclosure, not contradiction: the substitution check still passed at every sampled point -- the note names the lever that tightens the construction (measurement guide, item 10).

> Every notice raised on this run appears above, translated; nothing is withheld to an internal file. Estimates are marked as estimates.

4. What was not checked

• A second NUMERICAL method was not run on this problem; the exact energy-

quadrature relation printed in section 3 is the independent (forward- error) check on this run.

• No structural invariant was detected by the invariant search; the exact

relation printed in section 3 is the conserved-quantity audit of this run.

• Residuals are sampled at a finite set of points. They are strong evidence,

not a proof valid at every point of the interval.

• Nonlinear problems may admit more than one solution. This report covers

the solution reported here and makes no claim that it is the only one.

5. Scope

This report covers the problem exactly as stated in section 1. It makes no claim regarding the accuracy of that statement as a model of any physical system, nor regarding fitness for any particular purpose. Selection and validation of the model remain the responsibility of the client.

6. Issued and bound

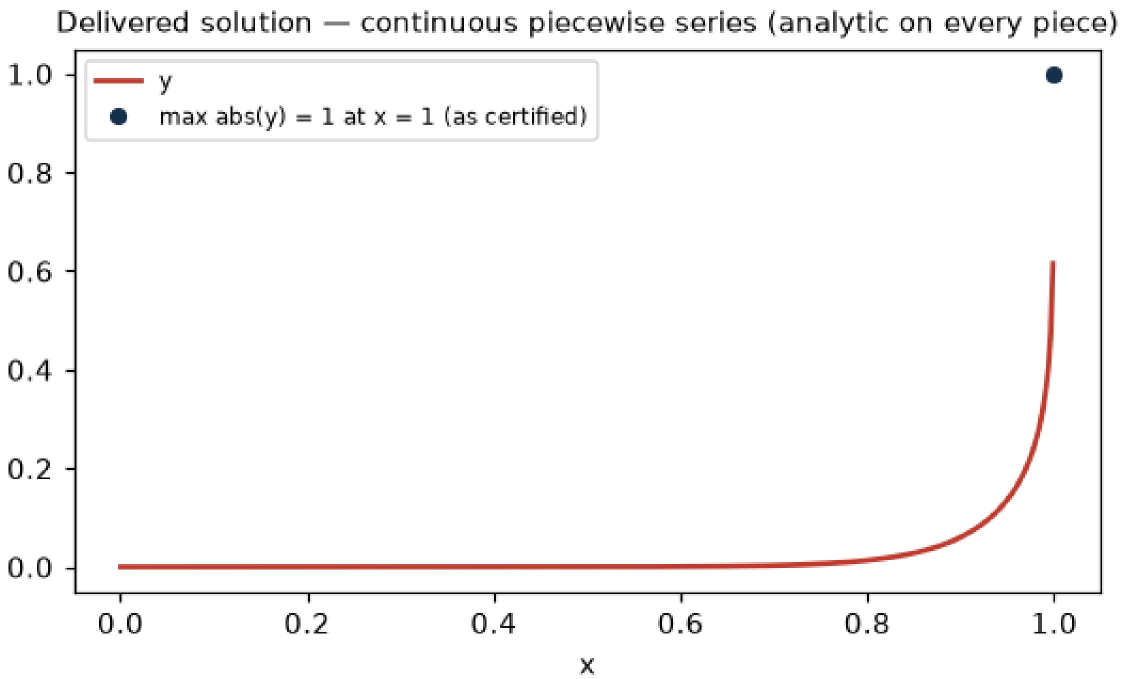
Prepared for	Acme Labs
Date	2026-08-29
Reference	LQI-2026-Troesch-15
Prepared by	Luminary Quantum Institute, LLC

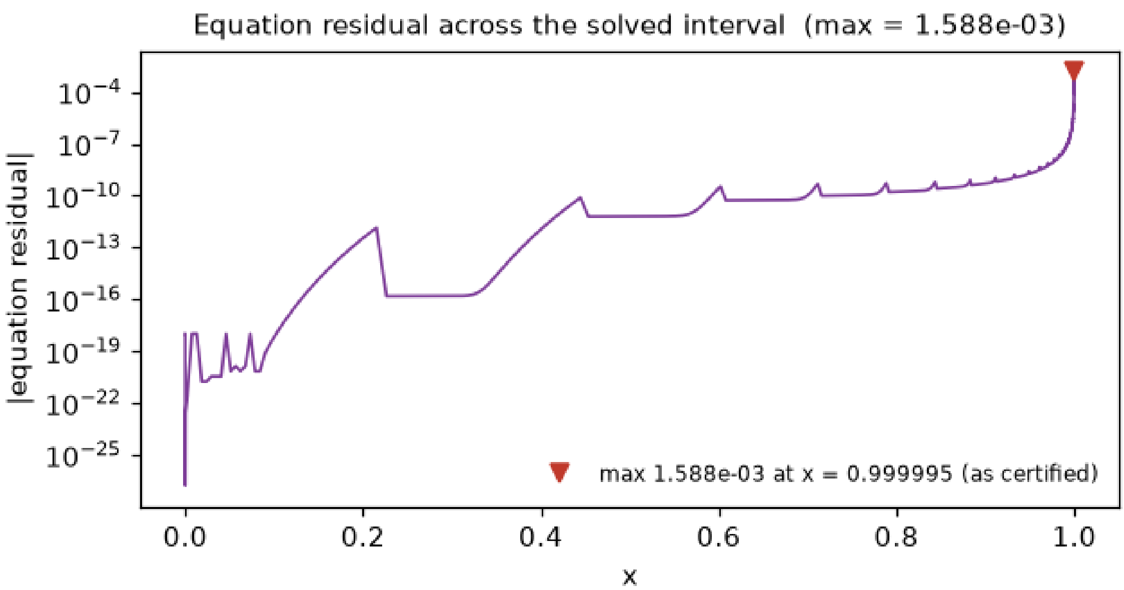
Prepared for	Acme Labs
Tool	SIRPY 0.59.15
Report format	v1.2
Run environment	numpy 2.5.2, scipy 1.18.1, sympy 1.14.0, mpmath 1.3.0, reportlab 5.0.1; IEEE-754 float64 throughout; the procedure is deterministic (no random seeds)
Verification procedure	substitution of the solution into the stated equations; exact differentiation of the solution representation; sampling at interior points off the interface nodes; thresholds as stated in section 3
Reproduction	requires only the artifact and the stated equations; Provider software is not required
Solution artifact	LQI-2026-Troesch-15-T3.artifact.json
Artifact SHA-256	ce472044612b41fc56e71cb744ee6f1e92d99cd9303a4b79c1f3dfbb1ac3853c
Hash procedure	SHA-256 of the artifact file exactly as delivered, byte for byte. Linux/Mac: sha256sum <file>. Windows: certutil -hashfile <file> SHA256 or PowerShell Get-FileHash <file>. Any OS: python -c "import hashlib;print(hashlib.sha256(open('<file>','rb').read()).hexdigest())". Do NOT load and re-serialize the JSON - a recomputed serialization will not match, by design; the delivered bytes are the binding. A renamed file with a matching hash is fine; the hash, not the name, binds.
Governing terms	NONE — SAMPLE, not issued

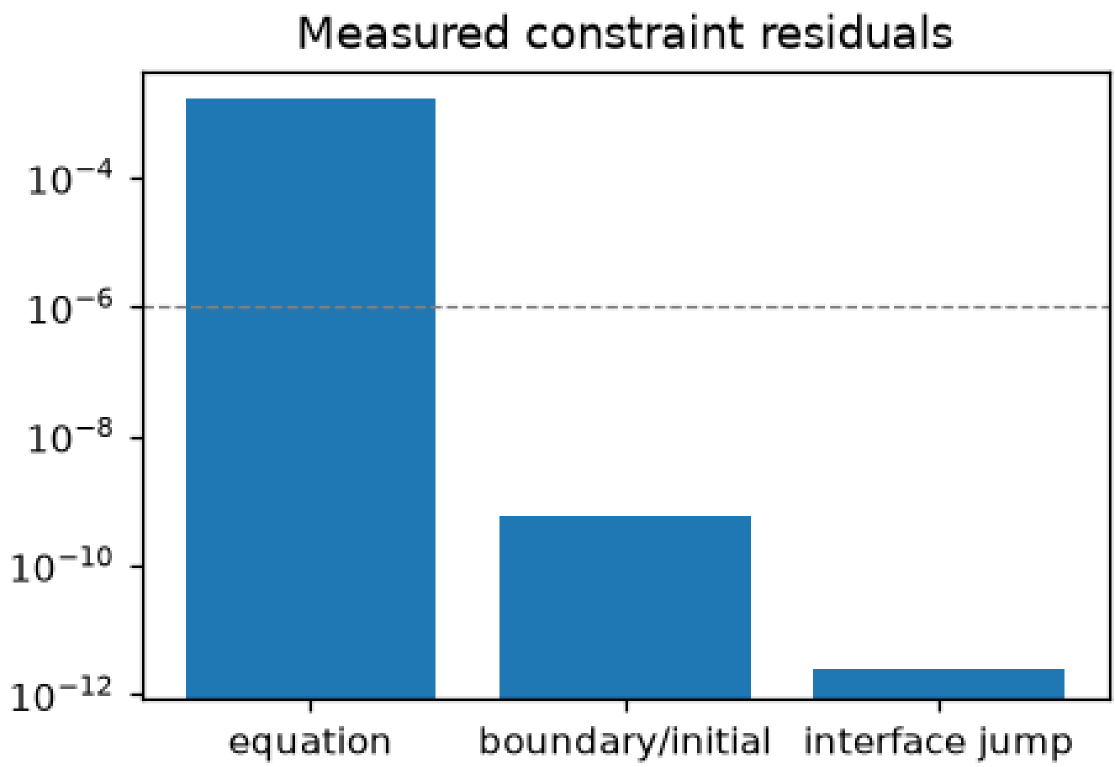
7. Reproducibility and Correction Policy

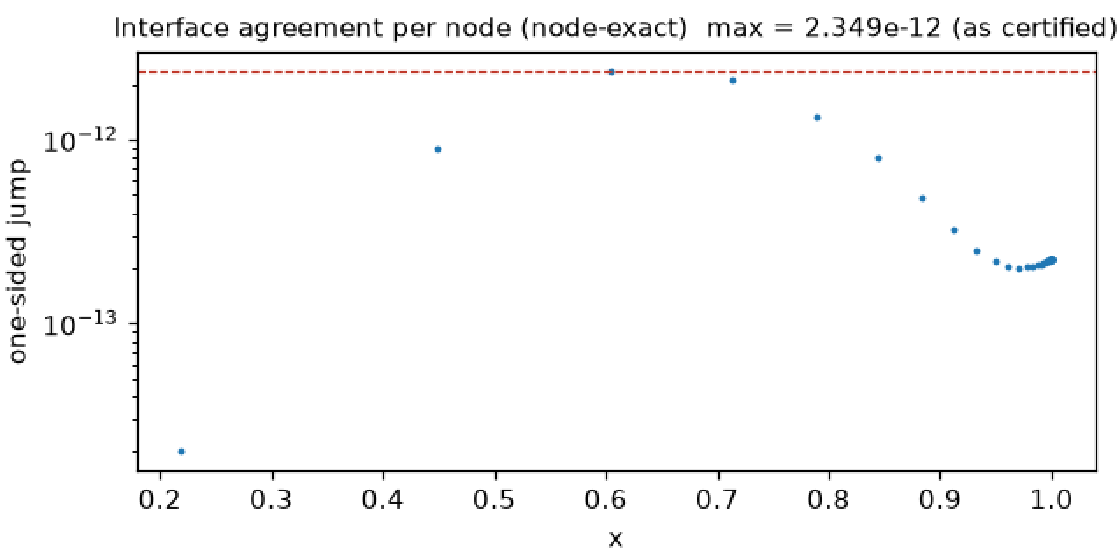
This Report is limited to the computational reproducibility of the results identified in it. The Report identifies the applicable input specification, solution artifact and artifact hash, software version, verification procedure, and the stated numerical tolerance. For ninety (90) days after delivery, the contracting customer may submit written notice identifying a specific reported result that the customer contends does not reproduce under the procedure stated in this Report, together with the information reasonably necessary to evaluate the claim. A Covered Reproducibility Failure exists only if the identified result cannot be reproduced within the stated tolerance using the identified inputs, solution artifact, software version, and procedure, and the failure is not caused by altered or incomplete inputs, third-party data, a changed software or hardware environment, corrupted files, or use outside the stated scope. If a Covered Reproducibility Failure is established, Provider will, at its option, either correct and reissue the affected Report at no additional charge or, if Provider cannot complete the correction within thirty (30) days after receiving the information reasonably necessary to perform the correction, refund the fees actually paid for the affected Report. Correction or refund under this policy is the customer's sole and exclusive remedy for a Covered Reproducibility Failure, and Provider's aggregate liability arising out of or relating to the affected Report will not exceed the fees actually paid for that Report. This Report does not constitute a representation that the customer's equations, model, assumptions, inputs, or methodology are suitable for any particular purpose, or that any decision based on the Report will produce a particular result. SAMPLE REPORT: no Engagement Terms are incorporated and no commitment is made; this section shows the policy language that governs issued reports.

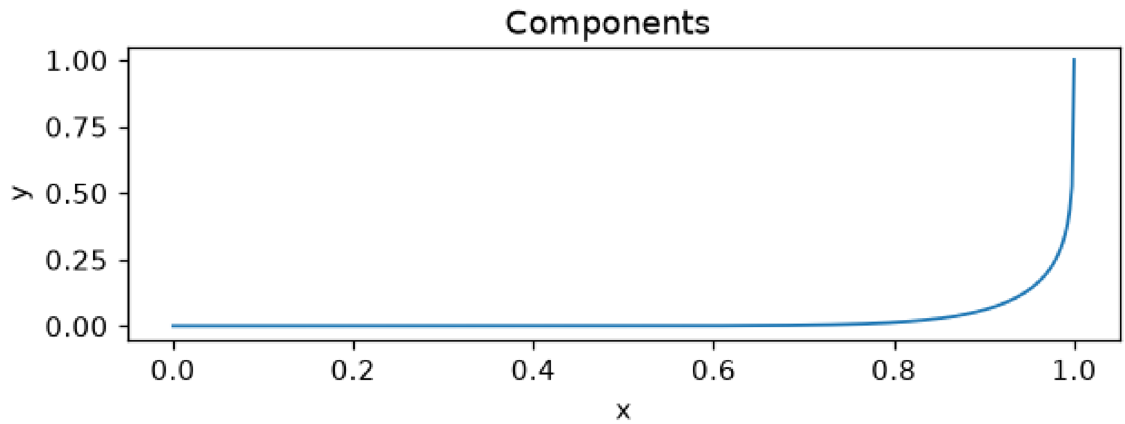
Delivered figures

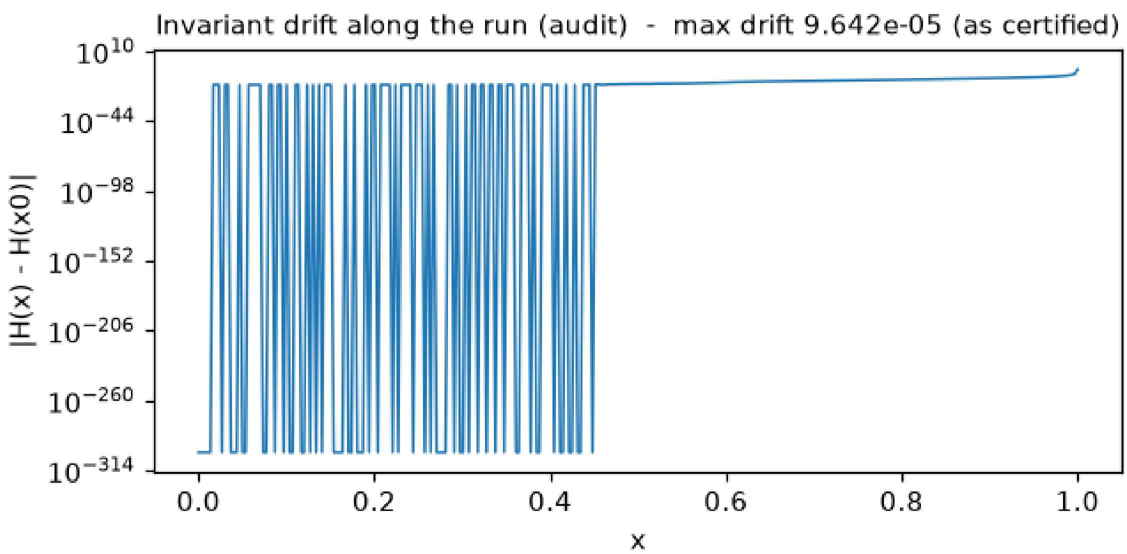

Delivered solution — continuous piecewise series (analytic on every piece)
y
max abs(y) = 1 at x = 1 (as certified)
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
x

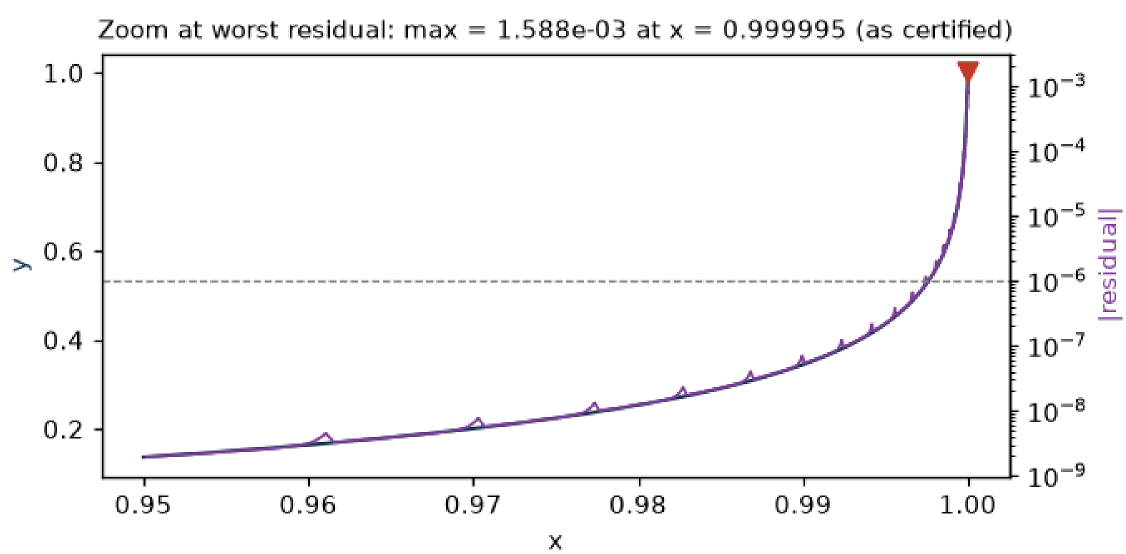

Equation residual across the solved interval (max = 1.588e-03)
|equation residual|
10^{-4}
10^{-7}
10^{-10}
10^{-13}
10^{-16}
10^{-19}
10^{-22}
10^{-25}
max 1.588e-03 at x = 0.999995 (as certified)
0.0
0.2
0.4
0.6
0.8
1.0
x


Measured constraint residuals
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
equation
boundary/initial
interface jump


Interface agreement per node (node-exact) max = 2.349e-12 (as certified)
one-sided jump
10^{-12}
10^{-13}
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
x


Components
y
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
x


Invariant drift along the run (audit) - max drift 9.642e-05 (as certified)
|H(x) - H(x0)|
10^{10}
10^{-44}
10^{-98}
10^{-152}
10^{-206}
10^{-260}
10^{-314}
0.0
0.2
0.4
0.6
0.8
1.0
x


Zoom at worst residual: max = 1.588e-03 at x = 0.999995 (as certified)
y
1.0
0.8
0.6
0.4
0.2
|residual|
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
0.95
0.96
0.97
0.98
0.99
1.00
x

Premium analytics

Premium convergence analytics, conditioning indicators, perturbation sensitivity and the LaTeX recurrence export are delivered in the accompanying **Premium Addendum** (`LQI-2026-Troesch-15-T3.addendum.pdf`), included in this package.

Issued by LQI. The order manifest (ORDER_AND_DELIVERY.txt) is included in this delivery package alongside this document and the artifact; the terms of service applicable to this order are those supplied with your order confirmation and named by version in the delivery manifest.